

Data Analysis and Visualization using excel

Assignment

1. Explain the 5 V's of Data. Briefly describe each with one example.
2. What is an API? Explain how APIs are used for data collection with a suitable example.
3. What is data streaming? Explain how streaming differs from batch processing with an example.
4. What is data extraction? Explain its role in the data life cycle with an example.
5. Explain the steps to import data into Excel from an external file (CSV / Text / Database).
6. What are Pivot Tables and Slicers in Excel? Explain how they help in data analysis.
7. Differentiate between quantitative and qualitative data. Explain their types with suitable examples.
8. Explain some real-world scenario where data visualization is used and justify its importance.
9. What is the need of data storage. Why are **relational databases** still widely used in organizations?
10. Why is data cleaning necessary before data visualization? Explain how duplicate data and missing values can affect data analysis and the accuracy of visualizations.